Example

Router (config) # ip route < destination - network > < subnet mask > < next hop - IP >

command prop / Router CI:

Router > enable

Router # configure terminal

Router config # ip route 192.168.20 255.255.

Router config # end.

Router # show ip route.

Sample output:

15. 192.168.20/24 [1/0] via 10.0.0.2.

Exp: 11 Routing at the network layer

(a) Static Routing protocol configuration using Cisco packet Tracer.

AIM: TO configure static routing b/w different network using cisco packet Tracer and verify packet.

~~Alg~~ Algorithm:

1) Open cisco packet Tracer and place two routers switches and PC,

2. connect the pcs to them respective switches and routers

3) Assign IP address and subnet masks to the pc and route interface

4) Configure the IP address of the router interface and enable them.

5) Verify the routing table [Router # Show ip route]

6. Use ping to test communication b/w pcs on different networks

7) Observe packet transmission in simulation mode

RIP Routing

Router > enable

Router configure terminal

Router config t router rip

Router (config-router) # network 192.168

Router (config-router) # end

Router show ip route

Sample output

R 192.168.22 [24/120] via 10.0.2

(11)(b). RIP configuration using Packet Tracer.

AIM:

To configure the Routing information protocol (RIP) and verify dynamic routing between different network using cisco Packet Tracer.

Algorithm:

1) Open cisco packet Tracer and create a topology with multiple routers and networks

2) connect the routers and devices using suitable cables

3) Assign IP addresses and subnet masks to all devices and router interfaces

4) Enter RIP configuration mode on each router

5) verify the routing table using show ip route.

6) Configure the appropriate network every router

7) Test connectivity b/w pcs in different network using RIP

8) Enter RIP configuration mode on each router,

Result:

RIP was configured successfully. The routers exchanged routing information dynamically learned remote route and forward packets b/w the networks